

Bachelor's Thesis

Seq2Seq Mandarin-to-Cantonese Style Transfer: Colloquial Subtitle Translation in Hong Kong Urban Cinema

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Abstract

Cantonese, despite its large number of speakers worldwide, remains under-resourced in the field of natural language processing (NLP). This study presents a Mandarin–Cantonese parallel corpus constructed from subtitles of four Hong Kong Cantonese-language films and fine-tunes the mT5-small model for Mandarin-to-Cantonese machine translation in the context of film dialogue. The dataset was manually annotated to ensure alignment between Mandarin subtitles and natural Cantonese dialogue, and further expanded through paraphrasing-based data augmentation. The model was fine-tuned on the augmented training set, comprising 9,948 instances from 5,974 subtitle groups, and evaluated using automatic metrics and error analysis. The results show that the fine-tuned mT5-small model achieved 28.66 BLEU, 24.12 chrF, and 19.13 chrF++, underperforming compared to the Google Translate baseline. While the model is able to learn basic lexical and syntactic correspondences between Mandarin and Cantonese, the generated outputs often lack the naturalness and colloquial features characteristic of spoken Cantonese. This study highlights the challenges of modelling spoken language varieties and provides a preliminary dataset and baseline for future research in Mandarin-to-Cantonese subtitle conversion.

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1 Introduction

In discussions of the Chinese language (中文/汉语/华文), it is often erroneously assumed to be synonymous with Mandarin (官话/普通话/國語/华语), the standardised variety officially defined and promoted by the governments in Mainland China, Taiwan, and Singapore. This conflation, however, overlooks the linguistic diversity encompassed by the term "Chinese." As Wang and Sun (2015) pointed out, it is more appropriate to regard Chinese as a language family rather than a single language. Apart from Mandarin, the Sinitic family comprises several major branches, including Xiang, Gan, Min, and Yue. Despite Mandarin's dominant official status, Yue Chinese also holds considerable prestige, particularly through its most representative variety, Cantonese (Li et al., 2019; Wu, 2024).

Cantonese originated in Guangzhou and the surrounding Pearl River Delta region in southern China (Snow, 2004) and preserves certain phonological features of Old Chinese (Tang, 2018). As a result, many classical Chinese poems rhyme naturally in Cantonese but not in Mandarin. Mandarin, spoken primarily in northern China, had already been designated the national language of the Republic of China in 1932 (Chen, 2023), when the Republic of China governed Mainland China, well before the founding of the People's Republic of China in 1949. The PRC later strengthened the promotion of Mandarin through the 2001 Law on the Standard Spoken and Written Chinese Language (Ministry of Education of the People's Republic of China, 2015). Within this policy environment, the use of Cantonese in Guangdong and Guangxi has shown signs of decline, contributing to movements such as the "Protecting Cantonese Movement" in Guangzhou, which advocates for the preservation of Cantonese and its cultural roles (Li et al., 2022).

Meanwhile, due to its 156-year separation from Mainland China as a British colony, Hong Kong followed a divergent sociopolitical, economic, and cultural trajectory. According to Census and Statistics Department, Hong Kong (2023), 90.6% of the population aged five and above reported Cantonese as their usual spoken language, making it the de facto mother tongue of Hong Kong residents. Hong Kong is therefore the principal hub for the development, transmission, and preservation of Cantonese, both linguistically and culturally.

During the colonial period (1841–1997), the Hong Kong film industry gained significant global influence, facilitated by political stability and the territory's relatively high degree of freedom of speech in Asia at that time (Mak, 2018). The Hong Kong cinema has always operated within a cross-border and cross-cultural context (Hau & Shiraishi, 2013), where movies were redubbed or subtitled in other languages to be screened all over the world. To the general Chinese-speaking communities, in particular the overseas Cantonese diasporas, Hong Kong's Cantonese-speaking movies serve as a key cultural bridge, sustaining linguistic continuity and contributing to the shaping and preservation of Cantonese cultural identity (Abbas, 1997).

Despite its cultural significance, Cantonese remains as low-resourced language in the contexts of education and Natural Language Processing (NLP) (Bauer, 2016). Unlike Mandarin, Cantonese as a colloquial language lacks standardised orthography of Cantonese-specific characters (Bauer, 2018). Furthermore, Hong Kong has no official curriculum, textbooks, and proficiency examinations for Cantonese L2 learners. For Cantonese L1 speakers, literacy is developed in Standard Chinese (Mandarin) through traditional characters; written Cantonese is acquired "indirectly through exposure to its pervasive use" (Bauer, 2018) outside of the classroom. As a result, frequent exposure to media such as verbatim subtitles would be helpful for L2 learners' language acquisition.

However, as a common practice in Hong Kong, a majority of movie subtitles are written in Standard Chinese. These subtitles commonly paraphrase or generalise the Cantonese dialogue, filtering out Cantonese-specific vocabulary, expressions, and grammatical constructions. For learners who rely on subtitles to comprehend spoken dialogue, this mismatch can hinder acquisition and obscure the linguistic features unique to Cantonese. Constructing Cantonese-written subtitles would therefore benefit Cantonese L2 learners and expand the availability of Cantonese linguistic data.

In light of this gap, this study explores the feasibility of converting subtitles from written Mandarin in Simplified Chinese into natural Cantonese in Traditional Chinese. Rather than applying explicit style labels, genre-diverse training data is used to expose the model to a broader range of stylistic variation than a single-genre corpus would provide. By training on film dialogue spanning contrasting registers, the model encounters variation reflected in distinct lexical choices, pragmatic constructions, and patterns of sentence-final particle usage.

2 Related Work

2.1 Mandarin-Cantonese Machine Translation

A majority of related studies share the opinion that Mandarin and Cantonese as Chinese dialects are genetically related despite being mutually unintelligible. Recent work on Mandarin-Cantonese Machine Translation (MT) has focused on utilising limited data through semi-supervised and unsupervised methods

Zhang (1998) proposed one of the earliest frameworks for dialect machine translation within the Chinese language, adopting a knowledge-based approach that integrates linguistic rules, a word-collection list, and a dialect dictionary. In this system, Pinyin input is processed through three main components: (i) syntax conversion according to the target dialect when necessary, (ii) word-sense disambiguation using hierarchical phrase analysis and word-collection processing, and (iii) lexical substitution into the target variety. However, no subsequent system development or empirical evaluation was reported, thus the approach remained conceptual.

Wong and Lee (2018) also proposed a knowledge-based approach for Mandarin-to-Cantonese translation, introducing explicit modelling of register variation. Since lexical items in Cantonese can vary greatly depending on the register, different translations are thus needed to correctly convey the meaning. Cantonese-unique lexical items tend to occur in low-register (casual/informal) Cantonese, where high-register (formal) Cantonese shares more lexical items with Mandarin. This system first uses the Stanford Chinese parser for word segmentation, part-of-speech tagging and dependency parsing, then applied forward maximal matching to Mandarin-to-Cantonese lexical mapping, conditioned with POS tag and type of register, plus syntactic transformation before outputting. As a result, with Translation Error Rate (TER) as evaluation metric, two models in accordance with high and low register outperformed the baseline, to the respective test-sets.

Supervised MT requires a considerable amount of parallel data. Cantonese corpora are comparatively scarce, not to mention parallel corpora. In response to this limitation, Wan et al. (2020) explored Chinese dialect MT in an unsupervised setting. Their experiment leveraged both commonality and diversity between dialects through pivot private embeddings, parameter sharing, and layer coordination. Two monolingual corpora for Mandarin and Cantonese, each containing 20 million sentences, were collected to train the model. The results surpassed the rule based baseline at the character level. In a similar vein, Dare et al. (2023) investigated Mandarin-to-Cantonese MT using an unsupervised approach. Their system takes input in either dialect, passes it through cross lingual embeddings and a shared encoder, and generates output using the target dialect’s dedicated decoder. Their findings show that a Transformer model with character tokenization and concatenated embeddings achieves the best character level BLEU score.

2.2 Low-Resource Neural Machine Translation

According to Ranathunga et al. (2023), there are an increasing number of Neural Machine Translation (NMT) researches of specific low-resource language pairs in recent years. A language pair is considered as low-resource when the size of available parallel corpora is below 0.5 million, and extremely low-resource under the size of 0.1 million. Traditional architecture of NMT relies on a bulk of parallel datasets, which is the major challenge in low-resource NMT. Despite data augmentation, semi-supervised, and unsupervised techniques as common solutions involved, there are three more approaches, including multi-NMT models that translate between more than one language pair within a model, transfer-learning using a NMT model trained on a high-resource language pair, and zero-shot NMT that produces promising translation with no parallel data for extremely low-resourced cases. The following papers explored the field of low-resource NMT in the case of translating from Mandarin to written Cantonese.

Liu (2022) presented the first benchmark of several NMT systems for translating from

Mandarin written in Traditional Chinese to Cantonese. The author adopted parallel sentence mining (PSM) to augment sentence pairs 35-fold, from 1,002 to 35,877. The study showed that a bidirectional LSTM model with PSM-based data augmentation was the best-performing combination, achieving a BLEU score of 12.37 at word level. Mak and Lee (2021) obtained 72K semantically similar parallel sentences from Mandarin and Cantonese Wikipedia respectively using the multilingual sentence embedding toolkit LASER, and trained an Encoder-Decoder model on this data. Compared to the baseline and the Baidu Fanyi translation tool, their system achieved the highest BLEU score on six out of eight datasets.

2.3 Text Style Transfer

Text Style Transfer (TST) is a sub-task of Natural Language Generation that "manipulates text attributes while preserving style-independent content" (Mukherjee & Dušek, 2024). Commonly studied attributes include formality, politeness, humour, toxicity, gender, etc. Despite substantial recent progress, TST remains a challenging research area. Prior works (Fu et al., 2018; Mukherjee & Dušek, 2024) highlight three major challenges of TST in unison: 1) the scarcity of parallel datasets or corpora in a particular style, 2) the absence of standard automatic evaluation metrics, and 3) the difficulty of separating style from content, which is a problem remained open in the research of TST.

Given that parallel style-aligned datasets are labour-intensive to construct and thus scarce, the dominant research directions centre around unsupervised and semi-supervised methods. He et al. (2020) presented a generative probabilistic model for unsupervised TST, which treats non-parallel data from two domains as partially observed parallel corpora, achieving substantial improvements over previous unsupervised methods on multiple style-transfer tasks. In the context of Chinese TST, Chou et al. (2025) combined BERT with a CycleGAN framework to transfer between the stylistic domains of Eastern Wuxia and Western fantasy, reporting decent performance on PPL and BLEU metrics. For the semi-supervised approach, Shang et al. (2019) proposed a semi-supervised TST model that leverages both small-scale parallel data and large-scale non-parallel data, introducing a cross-projection mechanism in latent space to effectively coordinate between the two types of data.

In contrast, research on supervised TST is barely developed. There are a few related parallel datasets, including GYAFC (Rao & Tetreault, 2018) and PASTEL (Kang et al., 2019), providing parallel sentences suitable for supervised TST. The scarcity of such corpora highlights the importance of constructing additional parallel datasets to facilitate research in this area.

3 Linguistic Background

Although Mandarin and Cantonese are both Sinitic languages, they are largely mutually unintelligible (Dalte, 2020; Lo et al., 2000; Wong & Lee, 2018; Zhang, 1998). The functional intelligibility experiments produced by Tang and Van Heuven (2009) confirmed this asymmetry: Mandarin speakers barely understand Cantonese, whereas Cantonese speakers achieve only limited comprehension of Mandarin. Accordingly, Mandarin and Cantonese constitute one of the most distant pairs among the 15 major Chinese dialects.

This section surveys several linguistic differences between Mandarin and Cantonese that are directly relevant to the challenges addressed in this study. It does not aim to provide an exhaustive comparison, but rather to illustrate contrasts that exist between the two dialects.

3.1 Orthography

The modern Chinese writing system consists of two different scripts: Traditional Chinese (TC) and Simplified Chinese (SC). TC characters refer to the historical forms evolved through thousands of years, and are still widely used in Hong Kong, Macau, and Taiwan, while SC characters are used in Mainland China, Singapore, and Malaysia.

SC was officially introduced by the Mainland Chinese government in the 1950s, following decades of discussion on script reform since the late Qing dynasty (DeFrancis, 1984). From then until the present, simplification efforts have been codified in official character tables, most recently in the Table of General Standard Chinese Characters (通用规范汉字表) published by State Language Commission (2013), which standardises 8,105 simplified forms in use. Simplification was achieved through several methods: reducing the number of strokes, adopting forms from cursive script, and merging multiple traditional characters into a single simplified form when they were homophonous. This merging caused a one-to-many mapping from simplified to traditional characters. As a result, SC is generally graphically less complex than TC.

To illustrate the visual differences, the following paragraph is excerpted from the Chinese-written work *Depression in China* (Liu, 2009):

① written in TC	「抑鬱症」就是這樣一種文化現象。在中國的語境下，它曾經被稱為「神經衰弱」。隨着文化對精神性疾病歸因認識的改變和西方醫學研究及醫藥事業發展對中國的影響，「抑鬱症」這個名稱才逐漸被愈來愈多的中國人接受和使用。
② written in SC	「抑郁症」就是这样一种文化现象。在中国的语境下，它曾经被称为「神经衰弱」。随着文化对精神性疾病归因认识的改变和西方医学研究及医药事业发展对中国的影响，「抑郁症」这个名称才逐渐被愈来愈多的中国人接受和使用。

Aside from the script employed, the two passages are the same with 91 characters each, excluding punctuations. Of these, 58 characters are shared across two versions, while the other 33 have simplified forms. Four characters exhibiting the greatest reduction in stroke count are highlighted in colours, including 變 → 变 (-14 strokes), 歸 → 归 (-13 strokes), 識 → 识 (-12 strokes), and 鬱 → 郁 (-11 strokes) in TC. In addition, the SC character 发 corresponds to two distinct TC characters: 發 in this context and 髮.

For consistency in this study, Mandarin examples are represented in Simplified Chinese and Pinyin (romanised form of Mandarin), while Cantonese examples are written in Traditional Chinese and Jyutping (romanised form of Cantonese).

3.2 Phonetics & Phonology

The phonological distinctions between Mandarin and Cantonese account for differences in their pronunciation. As mentioned by Gao et al. (2000) Mandarin has 24 syllable initials and 37 finals, constructing 410 base syllables, while Cantonese has 20 syllable initials and 53 finals, composing 660 base syllables. Mandarin’s five tones yield 1,400 distinct tonal syllables, while Cantonese’s six tones yield 1,800.

	⑤	①	④	③	②	⑥	⑦	⑧						
Original line	老	公	呀	,	睡	不	着	有	甚	么	可	以	做	?
Pinyin	lao	gong	ya	,	shui	bu	zhao	you	shen	me	ke	yi	zuo	?
Mandarin IPA	lau	kʊŋ	ja	,	ʃwei	pu	t͡sʌu	jou	ʃən	mɤ	kʰɤ	i	t͡suɔ	?
Jyutping	lou	gung	aa	,	seoi	bat	zoek	jau	sam	mo	ho	ji	zou	?
Cantonese IPA	lou	kʊŋ	aː	,	sey	pət	t͡sœrk	jeu	səm	mɔː	hɔː	ji	t͡sou	?
Translation	Hubby, what can I do when I can't sleep?													

Color legend: Green = onset; Orange = rime (diphthong); Blue = rime (vowel); Pink = rime (with nasal coda); Red = rime (with stop coda).

Table 1: An example subtitle line (written in Mandarin and Simplified Chinese) from the 1992 Hong Kong comedy movie 家有喜事 *All’s Well, End’s Well*. Tonal notations omitted for readability.

As shown in Table 1, the Mandarin onsets /ʃ-/ and t͡s-/ of 睡 (①), 甚 (②), and 着 (③) do not exist in Cantonese. Conversely, the Cantonese pronunciations of 着 (③) and 不 (④) respectively have the stop codas /-k/ and /-t/, which do not exist in Mandarin. Additionally, comparing the IPA transcriptions, other characters including 呀 (⑤), 么 (⑥), 可 (⑦), and 以 (⑧) show particularly distinct pronunciations between Mandarin and Cantonese. Therefore, Mandarin and Cantonese sound different, wordplay based on pronunciation and rhyme is thus often not transferable between the two varieties.

3.3 Syntax

Both Mandarin and Cantonese generally follow the SVO word order, while syntactic differences exist between these two dialects. One well-known difference occurs in ditransitive clauses. In the first example in Mandarin 1a, the word follows is *verb* + *indirect object* + *direct object*, while in the respective example in Cantonese 1b, the word order follows *verb* + *direct object* + *indirect object*.

- (1) At [00:05:40], O 記三合會檔案 *The H.K. Triad* (1999):

a. Mandarin

给 我 三 张 黄 票
gěi wǒ sān zhāng huáng piào
give 1SG three CLF yellow/porn ticket
'give me three pornoshow tickets'

b. Cantonese

畀 三 张 黄 飞 我
bei² saam¹ zoeng¹ wong⁴ fei¹ ngo⁵
give three CLF yellow/porn ticket 1SG
'give me three pornoshow tickets'

In the following example, the Mandarin version 2a obligatorily employs the *bǎ*-construction. In this structure, the second occurrence of 我 (*me*) is introduced by the valence marker 把 and functions as the patient, encoding the message of disposal. As a result, the object is fronted before the verb, yielding an OV order rather than the canonical SVO order.

By contrast, the Cantonese counterpart 2b remains the SVO word order and lacks the morphosyntactic features associated with the Mandarin *bǎ*-construction. It adopts the "regard ... as" meaning expressed through a copular structure 當.....係, without invoking a disposal construction.

- (2) At [01:27:06], 家有囍事 *All's Well, End's Well* (1992):

a. Mandarin

你 追求 我 的时候 把 我 当作 宝
nǐ zhuīqiú wǒ de shíhòu bǎ wǒ dāngzuò bǎo
2SG pursue 1SG ATV time BA me treat.as treasure
'When you were pursuing me, you treated me as a treasure.'
ATV = attributive and possessive particle 的 *de*
BA = valence marker 把 *bǎ*

b. Cantonese

你 追 我 个时 当 我 係 宝
nei⁵ zeoi¹ ngo⁵ go²si⁴ dong¹ ngo⁵ hai⁶ bou²

2SG pursue 1SG that.time regard me as treasure

‘When you were pursuing me, you treated me as a treasure.’

These syntactic contrasts highlight how similar surface word orders in Mandarin and Cantonese can mask deeper differences in argument realisation.

3.4 Lexical Inventory

According to the corpus-based estimation by Ouyang (1993), one-third of the lexical items in Cantonese are absent in Mandarin. Wong and Lee (2018) also pointed out that low-register Cantonese (casual, informal speech) frequently features unique lexical items that are not utilised in Mandarin. The Table 2 illustrates this with noun phrase used to swear at a person as “useless”. Mandarin has one term, whereas Cantonese offers seven variants with differing intensity, some specifying the age or gender of the referent.

Mandarin			
#	Chinese	Pinyin	Literal meaning
①	废物	fèi wù	<i>useless object</i>
Cantonese			
#	Chinese	Jyutping	Literal meaning
①	垃圾	laap ⁶ saap ³	<i>rubbish / garbage</i>
②	廢柴	fai ³ caai ⁴	<i>useless wood</i>
③	廢人	fai ³ jan ⁴	<i>useless person</i>
④	廢青	fai ³ cing ¹	<i>useless teen</i>
⑤	廢中	fai ³ zung ¹	<i>useless middle-aged person</i>
⑥	廢老	fai ³ lou ⁵	<i>useless elderly person</i>
⑦	廢閻	fai ³ hai ¹	<i>useless cunt</i>

Table 2: Synonyms expressing *good-for-nothing* in Mandarin and Cantonese.

3.5 Sentence-Final Particles (SFPs)

Among the various Chinese varieties, Cantonese possesses a particularly rich inventory of sentence-final particles (SFPs) (Ding, 2013; Lai & Xue, 2025; Law, 1990; Simpson, 2014). According to Matthews and Yip (2013), Cantonese has approximately 30 primary SFP forms, whereas Mandarin has only around eight. In addition, many Cantonese SFPs may co-occur and form stacks, making SFPs a salient and defining feature of Cantonese speech.

The distribution of SFPs is closely correlated with speech context. Leung (1992) and Tang (2002) conducted separate surveys on the occurrence of SFPs in local Hong Kong radio and television programs. Both studies reported that SFPs occur frequently in naturally occurring spoken interaction and are more common in less serious or informal topics and communicative settings. As a result, SFPs are especially prevalent in genres that aim to represent everyday colloquial Cantonese, such as Hong Kong films.

Functionally, Cantonese SFPs encode a wide range of discourse-sensitive meanings related to speaker attitude (Simpson, 2014). These include the marking of speech-act types, affectivity, emotional colouring, and evidentiality (Matthews & Yip, 2013). Due to their context-dependent and intonation-sensitive nature, Chinese SFPs are widely acknowledged to be difficult to translate accurately into English.

Looking more closely at a frequent SFP *lo⁴* in Cantonese (3a) below. *lo⁴* usually conveys the meaning of self-evidentiality, and "very obvious" in a tone of annoyance, which the speaker thinks that the matter is very clear that no further explanation is needed (Tang, 2008). By changing tones, the sound *lo* has two other SFPs at the first and the third tones. Raising to the highest tone 1 from the lowest tone 4 (3b), Sybesma and Li (2007) consider *lo¹* is very similar to *lo⁴*, that *lo¹* indicates the speaker does not want to admit the truth carried in the speech. It is "more evasive" and less blunt. *lo³* does exist, however it is grammatically ill in this context. *lo³* can be found in sentences that express or emphasise the change of state. Additionally, Cantonese SFPs lack standardisation of written form, the morpheme *lo* as SFP can be written as 囉 (most commonly seen), 咯, and 囉.

(3) At [00:10:49], 流氓師表 *The Teacher Without Chalk* (2000):

a. Original utterance - Cantonese SFP *lo⁴*

你 唔 搭 我 咪 唔 食 囉
nei⁵ m⁴ daap⁶ ngo⁵ mai⁵ m⁴ sik⁶ lo⁴

2SG NEG place-hand 1SG then NEG eat LO4

'Well, obviously if you don't place your hand on my shoulder, I will stop smoking.'

b. Adapted to SFP *lo¹*

你 唔 搭 我 咪 唔 食 囉
nei⁵ m⁴ daap⁶ ngo⁵ mai⁵ m⁴ sik⁶ lo¹

2SG NEG place-hand 1SG then NEG eat LO1

'If you don't place your hand on my shoulder, I will stop smoking. (where speaker is reluctant to say so and admit the fact that the listener gets the upper hand in the situation)'

c. Adapted to SFP *lo³*

* 你 唔 搭 我 咪 唔 食 囉
* nei⁵ m⁴ daap⁶ ngo⁵ mai⁵ m⁴ sik⁶ lo³

* 2SG NEG place-hand 1SG then NEG eat LO3

The meaning of Cantonese SFPs is interpreted depending on various factors, including discourse context, epistemic stance of speaker (Chor, 2018; Law, 2004). A majority of Cantonese SFPs have no one-to-one equivalents in Mandarin.

4 Methodology

Given the linguistic differences between Mandarin and Cantonese discussed in the previous section, Mandarin-to-Cantonese subtitle conversion goes beyond literal translation. In addition to lexical and syntactic divergence, natural Cantonese dialogue exhibits systematic use of sentence-final particles, colloquial constructions, aspect markers, and discourse-driven omissions that are typically absent from written Mandarin subtitles. As a result, this task lies at the intersection of machine translation and a form of language adaptation.

This study investigates whether the existing pre-trained multilingual neural machine translation (NMT) model can be effectively adapted to this task using supervised parallel data. In particular, it examines whether such models are capable of learning systematic Mandarin-to-Cantonese transformations from subtitle pairs. This section describes the dataset construction process, fine-tuning setup, and evaluation setup used in the experiments.

4.1 Dataset Construction

At present, there is no publicly available Mandarin-Cantonese parallel corpus derived from movie subtitles. The closest existing resource is Lee’s Cantonese-to-Mandarin parallel corpus, which is not publicly available, contains over 75,000 characters (Lee, 2011). However, it consists of only television programs, including one episode of news broadcast and drama series respectively. To address this gap, a new parallel dataset was constructed specifically for the purpose of Mandarin-to-Cantonese subtitle conversion.

4.1.1 Data Sourcing

As a contribution to Cantonese NLP, this study constructs a high-quality Mandarin-Cantonese parallel dataset with human intervention. Given the scarcity of reliable Cantonese NLP resources, manual verification is employed to ensure the accuracy of each sentence pair. The dataset is derived from four Hong Kong Cantonese-speaking films released between the 1980s and 1990s.

Film selection follows three criteria: (1) genre relevance to the theme *Hong Kong Urban Cinema*, (2) availability of video clips online, and (3) the availability of Mandarin subtitles in Simplified Chinese SubRip Subtitle files (`.srt`) of the film on `OpenSubtitles`. Cantonese subtitle written in Traditional Chinese files, when available, are used as supplementary references to facilitate faster annotation and validation, but are not required for selection criteria. As a result, 7,468 individual lines of subtitle instances were collected to construct the dataset, from four movies (see Table 3).

Movies	Genre	Duration	Lines per film
最佳拍檔 <i>Aces Go Places (1982)</i>	Action, Comedy	93 mins	1,549
英雄本色 <i>A Better Tomorrow (1986)</i>	Action, Triad, Crime	95 mins	953
籠民 <i>Cageman (1992)</i>	Drama	145 mins	2,850
古惑仔 3 之隻手遮天 <i>Young & Dangerous III (1996)</i>	Action, Triad, Crime	98 mins	2,116

Table 3: Metadata of the four selected Hong Kong Cantonese-speaking movies for dataset construction.

4.1.2 Data Annotation

The data annotation process was carried out entirely by one single annotator who is a native speaker of Cantonese, grew up in Hong Kong writing the Hong Kong variant of Traditional Chinese, and is fluent in Mandarin with basic knowledge about writing Simplified Chinese. Subtitles of each movie are stored as one file in JSON Lines (`.jsonl`) format, where each subtitle pair is represented as an individual JSON object on a separate line. As illustrated in Listing 1, the task is formulated as a sequence-to-sequence transformation that maps Mandarin subtitles (tagged as `input`), which are translations of the original Cantonese dialogue, to the corresponding spoken Cantonese utterances as heard in the film (tagged as `target`), which serve as the gold-standard target. To compensate for the annotator’s limited familiarity with Simplified Chinese orthography, the orthographical converter `OpenCC` was utilised to convert between Simplified Chinese and the Hong Kong variant of Traditional Chinese.

```

{"input": " 现在跟你算账", "target": " 我而家同你計啊"}

{"input": " 就是你私吞帮会公款的那笔账", "target": " 係你穿櫃桶嗰條數"}

{"input": " 那八十万你怎么交代? ", "target": " 嗰八十萬你話點計啊? "}

{"input": " 对", "target": " 係呀"}

```

Listing 1: Four individual lines of subtitles extracted from the raw dataset of the movie *Young & Dangerous III* before any data processing.

The annotation procedure involved cross-checking the original movie audio against pre-existing Mandarin subtitles obtained from the previously mentioned website. For each subtitle pair, the annotator first evaluated whether the Mandarin subtitle accurately and naturally reflected the original Cantonese utterance. If the Mandarin subtitle was deemed acceptable, it was retained; otherwise, the annotator revised or retranslated the Mandarin line to better align with the spoken dialogue. When an utterance was audible in the movie but absent from the `.srt` file, it was manually added to the dataset.

This fidelity requirement is particularly important in the context of the present task. Since register and stylistic features are assumed to be at least partially encoded within individual line instances rather than through explicit labels, the Mandarin input must preserve the communicative intent, emotional tone, and social context of the original Cantonese utterance as much as possible. A Mandarin transcription that paraphrases or generalises the source dialogue risks obscuring these features, thereby limiting the model’s ability to learn such distinctions.

Regarding the annotation process, several linguistically motivated decision points are noteworthy. Firstly, sentence segmentation is decided based on a natural pause in a character’s speech, rather than segmenting an entire spoken turn as one single instance at once. This approach aims to improve the model’s learning efficiency and to produce more accurate sequence-to-sequence generation in downstream tasks. Secondly, lines spoken entirely in languages other than Cantonese are excluded from the dataset, while instances exhibiting Cantonese-English code-switching and commonly used English phrases in Cantonese are retained, as both are widely observed features of Hong Kong Cantonese (Snow, 2004). Although the four films also contain occasional lines in different Chinese dialects, English, and Dutch, these occurrences are rare and thus not included, in order to maintain focus on Mandarin-to-Cantonese subtitle conversion. Thirdly, lines involving Cantonese-specific wordplay, puns, or expressions without equivalents in Mandarin are excluded, as such instances cannot be faithfully rendered and thus would not provide meaningful training data. Finally, due to the non-standardisation of written Cantonese, a predefined standardisation list was used to guide transcription of Cantonese labels whenever a word had multiple possible forms. The full annotation scheme is provided in Appendix A.

4.1.3 Data Augmentation

Given the limited availability of labeled Cantonese data, data augmentation is necessary to mitigate overfitting and improve model generalisation. It increases both the quantity and diversity of training instances, thereby theoretically enhancing model robustness.

To this end, semantics-preserving paraphrasing was applied to the input side, while the Cantonese targets were kept unchanged. This design is motivated by the nature of the target data, which consists of literal transcriptions of spoken Cantonese dialogue. In addition, Cantonese sentence-final particles (SFPs) are highly sensitive to context and intonation and are pervasive in natural speech. Modifying the target side would therefore risk introducing unnatural or contextually inappropriate SFP usage.

Instead, paraphrasing was applied to the Mandarin inputs to reflect the many-to-one nature of the task, where multiple Mandarin surface forms may correspond to the same Cantonese utterance. This setup also mirrors real-world subtitle translation practices, in which the same spoken Cantonese line may be rendered into different Mandarin phrasings.

Following Wang et al. (2025), Chinese Large Language Models (LLMs) generally outperform Western LLMs on Mandarin tasks. Due to accessibility considerations, paraphrase generation was performed using the DeepSeek-V3.2 model via the DeepSeek interface, which produced one or more semantically equivalent Mandarin variants for each instance. A human-in-the-loop validation step was then conducted by the annotator. Candidate paraphrases were evaluated against the original film context and the corresponding Cantonese utterance, and only those that preserved semantic fidelity and appropriate register were retained. This step is necessary, as automatic paraphrase generation does not guarantee semantic accuracy or stylistic appropriateness.

```

{"input": "这几年我少吃了很多肉", "target": "呢幾年我少食好多肉類"}
{"input": "这几年我肉吃得少了", "target": "呢幾年我少食好多肉類"}

{"input": "闲来无事吃点健康食品", "target": "得閒無事食啲健康食品"}

```

Listing 2: Augmented subtitles extracted from the dataset of the movie *Young & Dangerous III*.

Accepted paraphrases were grouped together with their corresponding original instance, forming groups of semantically equivalent Mandarin inputs aligned to a single Cantonese target. These groups are separated by newline during preprocessing, which are later removed in downstream processing. Listing 2 shows two individual groups, with the first group augmented through paraphrasing.

As shown in Table 4, this approach increased the dataset size from 7,468 to 12,454 instances, representing an overall increase of approximately 60%. The augmentation rate for each movie ranges from 59% to 63%.

Movies	Before Data Augmentation	After Data Augmentation	Differences
AGP	1,549	2,596	+1,047
BTM	953	1,504	+551
CGM	2,850	4,746	+1,896
YND	2,116	3,608	+1,492
Total	7,468	12,454	+4,986

Table 4: Subtitle counts for each movie before and after data augmentation.

Data augmentation is typically applied after the train/dev/test split. However, in this study, augmentation was performed prior to splitting in order to maximise the amount of human-validated data available. The dataset was divided at the group level using an 80/10/10 split with a random seed of 42, where each group consists of one original subtitle pair and its augmented variants. This yields 5,974 groups in training set, 746 in dev set, and 748 in test set from a total of 7,468 groups. To prevent data leakage, augmented variants were excluded from the dev and test sets, retaining only the original

instance per group. As a result, the final training set contains 9,948 instances across the 5,974 groups, while the dev and test sets contain 746 and 748 instances respectively.

4.2 Fine-Tuning Setup

The Seq2Seq model selected for this study is the Google mT5-small (Xue et al., 2021), a multilingual variant of T5 trained on the mC4 dataset that covers 101 languages, including Chinese and English. Although the mC4 dataset does not explicitly differentiate between Chinese variants, manual inspection of the Chinese portion in the corpus reveals that both Mandarin and Cantonese instances are included. mT5-small was chosen over larger variants such as mT5-base due to computational constraints. The relatively small size of the training set further motivates this choice, as larger models may be more prone to overfitting on limited data.

Fine-tuning was performed using the HuggingFace **Transformers** library. The model was trained for five epochs with a batch size of eight, a learning rate of $2e-4$, and weight decay of 0.01. The AdamW optimizer was applied as the default optimizer of the **Seq2SeqTrainer**. Input and target sequences were tokenised and padded to a maximum length of 128 tokens, which is sufficient to cover the majority of subtitle lines in the dataset. The model was evaluated on the development set every 500 steps, and the best checkpoint based on development loss was retained for final inference. At inference time, beam search decoding with four beams was applied to generate Cantonese outputs.

Hyperparameter	Value
Model	google/mt5-small
Epochs	5
Batch size	8
Learning rate	$2e-4$
Weight decay	0.01
Max sequence length	128
Optimizer	AdamW
Decoding	Beam search (4 beams)
Best model selection	Development loss
Random seed	42

Table 5: Hyperparameter configuration for mT5-small fine-tuning.

The hyperparameter configuration is summarised in Table 5.

4.3 Evaluation

This study evaluates model performance using three automatic metrics, including BLEU (Papineni et al., 2002), chrF (Popović, 2015), and chrF++ (Popović, 2017). All metrics are computed using the sacreBLEU toolkit to ensure comparability.

Character-level tokenization is adopted via the **zh** tokenizer in sacreBLEU, which is more appropriate than word-level tokenization given the absence of explicit word segmenters in the Chinese language. BLEU measures n-gram precision between hypothesis and reference translations and remains the most widely used metric in MT research. On the other hand, chrF computes the F-score over character n-grams, while chrF++ additionally incorporates word n-grams, providing a complementary perspective on translation quality. Given the orthographic non-standardisation of written Cantonese discussed in the previous section, where multiple character forms may validly represent the same word, character-level metrics such as chrF and chrF++ offer greater robustness to surface variation than BLEU alone. All metrics are computed at corpus level across the 748 test instances.

It should be noted, however, that these automatic metrics evaluate surface-level n-gram overlap against a single reference translation, and are therefore unable to capture register or stylistic appropriateness. As discussed in Section 2.3, the absence of standard automatic evaluation metrics remains an open challenge in TST research (Fu et al., 2018; Mukherjee & Dušek, 2024). Consequently, whether the model outputs align with the expected colloquial register of the source film context cannot be fully assessed through automatic metrics alone, and this dimension of the task is acknowledged as a limitation of the current evaluation setup.

Despite these limitations, Google Translate is adopted as the baseline system for mapping from Chinese (Simplified) to Cantonese. This baseline serves as a point of comparison to contextualise the performance of the proposed model under realistic conditions. It should further be noted that Google Translate is trained on substantially larger and more diverse corpora and is not specifically optimised for the colloquial, genre-specific register targeted in this study. Its higher automatic metric scores therefore do not necessarily indicate superior performance on the style transfer dimension of the task.

5 Results

This section reports the performance of the mT5-small model (with and without data augmentation) and the baseline system on the test set containing 748 instances, evaluated against the three automatic metrics.

Systems	BLEU	chrF	chrF++
Google Translate baseline	37.62	31.83	29.25
mT5-small (with training set augmentation)	28.66	24.12	19.13
mT5-small (without training set augmentation)	27.47	23.43	18.60

Table 6: Automatic evaluation results on the test set (748 instances).

As shown in Table 6, the baseline system, Google Translate, outperforms the mT5-small model across all three evaluation metrics. In particular, the baseline achieves a BLEU score of 37.62, compared to 28.66 for mT5-small, indicating a greater n-gram

overlap for baseline with the reference translations. A similar trend is observed for chrF and chrF++, where the baseline consistently yields higher scores, suggesting better character-level alignment with the references. The largest gap is observed for chrF++, with a difference of 10.12 points between the two systems.

To assess the contribution of data augmentation, an ablation study was conducted by training mT5-small on the original training set only, while keeping the dev and test sets unchanged. The augmented model marginally outperforms the non-augmented model across all three metrics, with differences ranging from 0.53 to 1.19 points. These marginal gains suggest that simply increasing training data quantity through paraphrasing has limited impact on model performance at this scale. This points towards pre-training exposure to Cantonese as a more critical bottleneck than data augmentation alone, though further experiments, such as comparing models with varying augmentation rates or augmentation strategies, would be needed to confirm this interpretation.

It should be noted that these metrics primarily capture surface-level similarity between system outputs and reference translations. While chrF and chrF++ are well-suited for evaluating Chinese text due to their character-level formulation, and BLEU reflects n-gram precision, these metrics do not directly assess stylistic or pragmatic aspects of the generated Cantonese, such as the appropriateness of sentence-final particles or register.

6 Discussion

The results show that mT5-small fine-tuned on the augmented training set underperforms the Google Translate baseline across all three metrics. While this gap is considerable, it should be interpreted in context. A direct comparison with prior work is difficult due to differences in tokenisation, test sets, and evaluation methodology; however, Liu (2022), working on a closely related Mandarin-to-Cantonese NMT task, reported a best BLEU score of 13.22 using a bidirectional LSTM with Byte-Pair Encoding trained on 35,877 augmented pairs at word level. The substantially higher scores in the present study likely reflect both the advantages of multilingual pre-training in mT5 and the use of character-level evaluation, which tends to yield higher absolute scores than word-level BLEU. This suggests that mT5’s multilingual pre-training provides a meaningful starting point even under limited parallel data conditions.

That said, the performance gap with Google Translate is real and points to a fundamental asymmetry between the two systems. mT5-small is pre-trained on the mC4 corpus, whose Chinese portion contains only limited amounts of Cantonese, whereas Google Translate is trained on substantially larger and more diverse parallel corpora. As discussed in the following subsection, however, a closer look at the normalisation experiments suggests that a considerable portion of this gap stems from surface-level formatting artefacts rather than failures in semantic or syntactic generation.

6.1 Effect of Post-Processing Normalisation

A notable finding emerges from manual inspection of the model outputs, which reveals two systematic surface-level inconsistencies in the mT5-small predictions: punctuation form and orthographic script. Regarding punctuation, mT5-small consistently produces ASCII punctuation marks rather than their Chinese full-width equivalents. For instance, generating the English comma (”,”, U+002C) instead of the Chinese fullwidth comma (”, ”, U+FF0C). As illustrated in Table 7, although these character pairs are visually similar, they occupy distinct Unicode code points and are therefore treated as non-matching tokens by automatic evaluation metrics, artificially penalising the model’s scores. Regarding orthography, while Cantonese is conventionally written in Traditional Chinese, mT5-small occasionally produces Simplified Chinese characters in its output. A representative example is the word 開心 (*to feel happy*), which appears four times in the test set and is rendered as the Simplified form 开心 in all four instances by mT5-small, whereas the baseline correctly produces the Traditional form throughout.

Name	Character	Unicode
Fullwidth Comma	,	U+FF0C
Fullwidth Exclamation Mark	!	U+FF01
Fullwidth Question Mark	?	U+FF1F
Comma	,	U+002C
Exclamation Mark	!	U+0021
Question Mark	?	U+003F

Table 7: Comparison of Chinese full-width (red) and English ASCII (blue) punctuation characters with their respective Unicode code points.

To quantify the impact of these surface-level inconsistencies, two post-processing normalisation steps were applied independently and in combination: orthography normalisation, which converts Simplified Chinese characters in the model output to Traditional Chinese, and punctuation normalisation, which replaces ASCII punctuation with their Chinese full-width equivalents. As shown in Table 8, orthography normalisation alone yields only modest improvements across all three metrics, with gains ranging from 0.3 to nearly 0.6 points. Punctuation normalisation, by contrast, produces substantially larger improvements, most notably a gain of 6.6 points on chrF++. Applying both normalisations in combination yields the best overall performance, narrowing the gap with the Google Translate baseline, though the baseline continues to outperform mT5-small across all metrics.

Systems	BLEU	chrF	chrF++
Google Translate baseline	37.62	31.83	29.25
mT5-small	28.66	24.12	19.13
mT5-small (orthography normalisation only)	29.20	24.51	19.43
mT5-small (punctuation normalisation only)	32.66	27.36	25.73
mT5-small (both normalisations)	33.20	27.77	26.08

Table 8: Automatic evaluation results on the test set (748 instances), comparing the baseline system and mT5-small under different normalisation settings, including orthography normalisation, punctuation normalisation, and their combination.

These findings suggest that a considerable portion of mT5-small’s metric penalty is attributable to surface-level formatting inconsistencies rather than fundamental errors in semantic or syntactic generation. Punctuation-aware fine-tuning or the enforcement of Chinese punctuation in training targets could therefore meaningfully improve model performance without any architectural modifications.

6.2 Error Analysis

While automatic evaluation metrics provide a quantitative comparison between systems, they do not fully capture qualitative differences in linguistic naturalness and adequacy. To better understand the limitations of the proposed model, an error analysis was conducted on representative outputs. Two major error types were identified: (i) script conversion without language transfer, and (ii) semantic misalignment.

6.2.1 Script Conversion without Language Transfer

A recurring error type observed in the mT5-small outputs is the production of Traditional Chinese text that is structurally and lexically Mandarin, rather than genuine Cantonese. This phenomenon is referred to here as *pseudo-Cantonese*. In such cases, the model appears to have learned to convert the orthographic script from Simplified to Traditional Chinese, without acquiring the corresponding Cantonese vocabulary and grammatical structures. This behaviour is consistent with the composition of the mC4 pre-training corpus, whose Chinese portion contains only limited amounts of Cantonese, as well as with the relatively small size of the fine-tuning dataset.

As shown in Table 9, the model consistently retains Mandarin lexical items and structures instead of producing natural Cantonese equivalents. In instance (1), the model fails to produce the Cantonese equivalent of 拍照 (*to take a picture*), namely 影相, and instead retains the Mandarin verb 拍. While semantically interpretable, this results in an expression that is unnatural in spoken Cantonese. From instance (2), the model generates 打打殺殺 (*fighting and killing*), which is acceptable in Cantonese; however, the phrase 圖咩 is not idiomatic. The verb 圖, common in Mandarin with the meaning of *to covet/seek*, is not used in this way in Cantonese. Instead, the reference employs the Cantonese interrogative construction 為咩先? , which conveys both the intended

meaning and an appropriate discourse nuance. Notably, the model correctly maps 什么 to 咩, suggesting partial acquisition of Cantonese lexical items. In instance (3), the model outputs the formal term 海洛因 for *heroin*, which, although understandable, is not typical in colloquial Cantonese. The reference instead uses 白粉, a commonly used slang expression in Hong Kong. Alternatively, the transliteration 海洛英 may also be used in Cantonese, but remains less natural in informal spoken contexts.

#	Input(cmn)	Target(yue)	mT5-small output	Translation
(1)	拍个照, 拍个照	影相, 影相	拍個照, 拍個照	Take a picture, take a picture.
(2)	打打杀杀图什么?	打生打死為咩先?	打打殺殺圖咩?	What's the point of fighting and killing?
(3)	他身上有海洛因?	佢個身有白粉?	佢身上有海洛因?	Does he have heroin on him?

Table 9: Examples of script conversion without language transfer in mT5-small output.

These examples demonstrate that the model has not acquired sufficient knowledge of Cantonese-specific lexis and discourse patterns, instead defaulting to Mandarin vocabulary rendered in Traditional Chinese script.

6.2.2 Semantic Misalignment

A further category of errors consists of outputs that fail to preserve the meaning of the source input and exhibit mixed Mandarin–Cantonese forms, resulting in expressions that are not fully natural in Cantonese. These errors are particularly problematic in the context of subtitle-based language learning, as learners may incorrectly accept such outputs as valid Cantonese.

Three representative instances are shown in Table 10. In instance (4), the model incorrectly maps 订 (*to book*) to 買 (*to buy*), resulting in a shift in meaning from a hotel reservation to a purchase action. In instance (5), the verb 撒尿 (*to urinate*) is incorrectly rendered as 攞尿 (*to obtain urine*), which is semantically incoherent in Cantonese. In instance (6), the model misinterprets the Mandarin *bǎ*-construction (把-construction), leading to an incorrect agent–patient relationship, where the original object–verb structure is distorted in the output.

#		Input (cmn)	Target (yue)	mT5-small output
(4)	Text	酒店订好了没?	同我搵到酒店未?	酒店買咗冇啊?
	Translation	Have you booked the hotel?		Has the hotel been bought?
(5)	Text	我真撒尿了	我真係瀨尿	我真係攞尿
	Translation	I really peed.		I really obtained pee.
(6)	Text	这么好心, 把唐三给背上来了?	咁好死, 揸返唐三上嚟?	咁好心, 畀唐三背返嚟啊?
	Translation	How kind (of sb.) carrying Tong Sam up here?		How kind of Tong Sam carried (sb.) back?

Table 10: Examples of semantic misalignment in mT5-small output.

These examples indicate that, beyond lexical substitution errors, the model exhibits failures in preserving sentence-level semantics, leading to outputs that are fluent in form but incorrect in meaning.

Taken together, these findings suggest that mT5-small is capable of learning basic Mandarin-to-Cantonese mappings under limited data conditions, but falls short in capturing the lexical and pragmatic features that characterise natural spoken Cantonese. Both predominant failure modes point to insufficient Cantonese representation in the pre-training corpus as the primary bottleneck, rather than architectural or dataset size limitations alone.

7 Limitations

The present study is subject to several limitations that contribute to the suboptimal results.

One major limitation concerns the annotation process. The dataset was annotated by a single Cantonese-native speaker who is also fluent in Mandarin. While this ensures a certain level of linguistic competence, it may introduce bias. In particular, the annotator may not be fully familiar with colloquial expressions commonly used in Mainland Mandarin, which can affect the naturalness and accuracy of the source-side inputs. As a result, imperfections in the input translations may propagate to downstream tasks and negatively impact model performance.

Additionally, the overall size of the dataset remains relatively small. Although the data are carefully curated, the limited scale restricts the model’s ability to learn more sophisticated and nuanced mappings between Mandarin subtitles and natural Cantonese dialogue. This limitation is reflected in the model’s inconsistent performance across different instances. Moreover, the generated outputs are often insufficiently colloquial, lacking the vernacular quality expected in natural Cantonese speech. This further suggests that the model struggles to capture the pragmatic and sociolinguistic aspects of the language that are central to the style transfer objective of this study.

Another limitation lies in the choice and availability of pre-trained models. The study employs mT5-small, a general multilingual model that is not specifically optimised for Cantonese, but rather for high-resource written Chinese varieties such as Mandarin. As Cantonese is relatively underrepresented in large-scale training corpora, the model may lack sufficient exposure to its distinctive lexical, syntactic, and pragmatic features. This constraint limits the model’s ability to reliably distinguish between Mandarin and Cantonese, thereby reducing its effectiveness in generating natural and idiomatic Cantonese outputs, regardless of the quality of the training data.

A further limitation concerns the evaluation methodology. The automatic metrics (BLEU, chrF, and chrF++) employed measure surface-level n-gram overlap against a single reference translation and are therefore unable to capture stylistic or register-level appropriateness. Given that this study incorporates implicit style transfer through genre-diverse training data, a human evaluation component would be necessary to fully

assess whether the model outputs align with the expected colloquial register of the source film context. This represents an important gap in the current evaluation setup.

8 Future Work

Based on the limitations identified in this study, several directions for future research can be proposed.

To improve dataset quality, future work could involve multiple annotators with diverse linguistic backgrounds, incorporating inter-annotator agreement measures to reduce annotation bias. Expanding the dataset with a larger and more genre-diverse corpus would also enable the model to learn more nuanced mappings and better capture register-appropriate expressions across different social contexts.

On the modelling side, future work could explore models specifically adapted to Cantonese or spoken language generation, and integrate external resources such as lexicons of Cantonese colloquialisms to improve idiomatic output. Future work should also incorporate human evaluation, assessing outputs on dimensions such as naturalness and register appropriateness involving multiple native Cantonese speakers, with inter-rater agreement reported—providing a more complete picture of model performance than automatic metrics alone. Finally, multimodal approaches incorporating audiovisual cues from films could help the model better capture situational context and generate more register-appropriate dialogue.

9 Conclusion

This thesis investigates the conversion of Mandarin subtitles into natural Cantonese dialogue by fine-tuning the mT5-small model. To support this task, a parallel dataset was constructed from Hong Kong film subtitles, in which Mandarin inputs and Cantonese targets were manually curated according to the annotation guidelines. The model was trained on an augmented version of this dataset and evaluated using both automatic metrics and error analysis.

The results show that, while the system is able to learn basic lexical and syntactic correspondences between Mandarin and Cantonese, its outputs remain limited in naturalness and colloquial usage. In particular, the generated sentences often lack the vernacular quality expected in spoken Cantonese, indicating difficulties in capturing features beyond literal translation. Despite these limitations, the study demonstrates the feasibility of leveraging film subtitle data for Cantonese language modelling and highlights the challenges of modelling low-resource spoken language varieties using general-purpose multilingual models. Overall, this work provides a preliminary dataset and baseline system for Mandarin-to-Cantonese subtitle conversion and serves as a foundation for future research in this area.

A Appendix - Annotation Guidelines

These guidelines govern the construction of a Mandarin-to-Cantonese subtitle dataset drawn from Hong Kong Cantonese film audio. The goal is to produce natural, colloquial Cantonese dialogue that reflects authentic spoken usage as heard in Hong Kong films. The dataset is stored in `.jsonl` format with two fields: **target** for the transcribed Cantonese utterance, and **input** for the corresponding Mandarin translation.

Data Source & Transcription Principles

- Annotation is based primarily on the audible spoken dialogue in the selected film. Existing Cantonese subtitles may be used as a supplementary reference: adopt them if they are accurate, or correct them to match the audio if they are not.
- Include only clearly audible dialogue.
- Exclude all singing segments.
- Exclude lines that involve Cantonese-specific wordplay, puns, or expressions that have no equivalents in Mandarin.

Input & Target Format

- The **input** (Mandarin) column is written in Simplified Chinese using Mainland Chinese expressions and vocabulary.
- The **target** (Cantonese) column stores the literal spoken dialogue, written in Traditional Chinese following Hong Kong Cantonese orthographic conventions.
- Use Chinese full-width punctuation in both columns.

Segmentation & Alignment

- Segment utterances at natural speech boundaries, such as clause breaks, audible pauses, or breaths.
- Do not force a long stretch of dialogue into a single instance. Split it into shorter segments that each correspond to a natural unit of speech.
- Each **input** segment must align with its corresponding **target** segment in meaning and position.
- The **input** (Mandarin) must faithfully preserve the communicative intent and emotional tone of the original Cantonese utterance.

Lexical & Stylistic Conventions

- The word 阿sir is always written in lowercase in the **target** column, and should be consistently rendered as 警官 in the **input** column.
- Character names that appear in the dialogue are preserved as-is in both columns; they should not be translated or adapted.
- Include only lines spoken in Cantonese, Cantonese-English code-mixing, or English expressions that are frequently used in Cantonese contexts (e.g. *sorry*). Exclude any lines spoken in other dialects/languages.

- For iterative reduplication where the same character is repeated more than three times in the audio, transcribe it as a single instance of that character followed by “——！” (e.g. audio heard as 喂喂喂喂 is transcribed as 喂——！).
- Stuttered speech is marked using the format “x.....x”, where *x* is the stuttered syllable (e.g. 我.....我).
- For sentence-final particles, please follow the conventions documented at [Jyutping](#).

Character Usage Guidelines Cantonese written forms include several sets of characters that are easily confused or variably used across annotators. With reference to Education Bureau, Hong Kong (2026), Ministry of Education, Republic of China (2024), Research Centre for Humanities Computing, The Chinese University of Hong Kong (2014), and 石見田 (2014), tables below specify the correct character for each context. Refer to them whenever there is uncertainty.

Set of characters	Meaning & Examples
嘅 / 既	嘅 〈gei ³ 〉: 〈方〉①助詞, 的, 表示領有或修飾關係: 我嘅書 (我的書)。②語氣詞, 的, 一般用於陳述句的末尾表示強調: 噉做係咁嘅 (這樣做是對的)。 既 〈gei ³ 〉: 完成、已經: 既成定局。既來之則安之。
畀 / 俾 / 比	畀 〈bei ² 〉: ①動詞, 給; 給予: 分曹衛之田以畀宋人。 ②〈方〉介詞, 被: 畀人鬧 (被罵)。③〈方〉讓: 畀佢走 (讓他走)。 俾 〈bei ² 〉: 使, 使用; 讓。 比 〈bei ² 〉: ①比較。比試。②比賽雙方得分對比: 三比二。③相類。④比方。⑤比喻。⑥仿照。⑦比較且類數量的倍數關係。⑧介詞。
下 / 吓	下 〈haa ⁶ 〉: 作一切非句末助詞之用。 吓 〈haa ² 〉: 作句末助詞之用。
地 / 哋	地 〈dei ⁶ 〉: 用作助詞, 用在形容詞後: 甜甜地 (甜甜的)。 哋 〈dei ⁶ 〉: 們, 加於人稱代名詞後表示複數: 我哋 (我們)。
抖 / 抖	抖 〈tau ² 〉: ①呼吸; 喘息: 抖氣 (呼吸; 喘息)。②休息; 歇: 抖下先 (先歇一下)。早抖 (晚安)。抖涼 (乘涼)。 抖 〈dau ² 〉: ①顫動、哆嗦: 精神抖擻。②揭露: 把事情抖出來。
返 / 番	返 〈faan ¹ 〉: 返回、回來: 返嚟未啊? (回來了嗎?)。 番 〈faan ¹ 〉: ①量詞。②指外國或外族。
晒 / 晒 / 曬	晒 〈saai ³ 〉: ①同「曬」。②〈方〉借用作「完全」、「全」, 作後置狀語: 做晒嘞 (全做好了)。天黑晒 (天全黑了)。 ③〈方〉太, 表示強調: 唔該晒 (太麻煩了)。多謝晒 (太感謝了)。 晒 〈can ² 〉: 與「晒」及「曬」沒有關聯。 曬 〈dau ² 〉: ①為「晒」之異體字。②曝曬, 即暴露在陽光的照射下: 曬多啲太陽 (多曬點太陽)。

Set of characters (cont.)	Meaning & Examples (cont.)
使 / 洗 / 駛	使 <sai²>: ①派遣。②命令: 支使。使喚: 一句唔該使死人 (意指只用一聲「唔該」就老實不客氣麻煩別人代辦不同差事)。③讓。④致使。⑤導致: 使人高興。⑥假如: 假使。⑦用: 使用: 借嚟使使 (借來用用)。 洗 <sai²>: ①洗滌。洗衣服。洗車。②清除。③除去: 清洗。④比喻殺光或搶光: 洗劫。⑤攝影的顯影和定影: 沖洗。洗照片。⑥玩牌時把牌攙和整理: 洗牌。 駛 <sai²>: 行駛、駕駛: 安全駕駛最緊要 (安全駕駛最重要)。
裏 / 裡	裏 <lei⁵>: ①裏頭。裏應外合。②手裏。房子裏: 裏面得個橙 (裡面只有一個橘子)。③這裏。哪裏。 裡 <zoek⁶>: 為「裏」之異體字, 為台灣教育部所頒布之正體字。
着 / 著	着 <zoek⁶>: ①接觸, 挨上: 着陸。②使接觸或附於別的事物上: 着色。不着痕跡。③用, 下: 着手。着力。着眼。④下落: 着落。遍尋無着。⑤派遣, 使令: 着人前來。着即施行。⑥助詞, 表示動作的持續等: 拿着。擺着。想着。⑦受到: 着涼。着魔。⑧燃燒, (燈) 亮: 着火。燈着了。⑨入睡: 睡着。⑩表示達到目的或有了結果: 猜着。打着。 著 <zoek⁶>: ①受到。②染上, 今作着: 著涼, 為台灣教育部所頒布之正體字。
冇 / 無	冇 <mou⁵>: ①動詞, 沒; 沒有: 無: 冇益 (沒有好處)。②副詞, 沒; 沒有: 冇嚟 (沒來)。③副詞, 不 (只有少數例子, 多數用「唔」), 表示否定: 冇錯 (好)。 無 <faan¹>: ①沒有。②不: 無論。③不論: 事無大小。④同「毋」。
搞 / 攞	搞 <gaau²>: 做、幹: 搞咩啊? (幹什麼?)。 攞 <gaau²>: 擾亂、打擾: 唔滾攞你啦 (不打擾你了)。
鍾 / 鐘	鍾 <zung¹>: ① (情感等) 集中、專一: 鍾意 (喜歡)。③姓。 鐘 <zung¹>: ①金屬製成的響器, 中空, 敲時發出響聲: 警鐘。鐘聲。②計時的器具: 鬧鐘。③〈方〉指鐘點: 一個鐘 (一小時)。夠鐘 (到點了)。
少 / 小	少 <siu²>: ①數量小 (跟「多」相對): 少數。少量。②缺少 (跟「多」相對): 凶多吉少。③丟失。④遺失: 屋裏少了東西。 小 <siu²>: 物之微也: 小菜一碟。雷聲大雨點小。
喺 / 係	喺 <hai²>: ①動詞, 在: 喺度 (在這裡)。喺屋企 (在家裏)。②介詞, 在; 從: 坐喺個度 (坐在那裏)。喺邊度嚟? (從哪裏來?)。 係 <hai⁶>: 系動詞, 作粵語的「是」之用: 你係邊位? (你是誰?)。
嚟 / 黎	嚟 <lai⁴>: 來: 行嚟行去 (走來走去)。嚟廣州 (來廣州)。喺邊度嚟 (從哪裏來)。嚟晒 (全來了)。 黎 <lai⁴>: 姓。

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